

1. Administrative & Study Identification

Full Study Title: A Phase 2, Randomized, Double-Blind Trial to Evaluate the Safety, Tolerability, and Immunogenicity of a Multivalent Pneumococcal Conjugate Vaccine in Healthy Infants
Short Title/Acronym: A Study to Learn About How a New Pneumococcal Vaccine Works in Infants
Protocol Number: C4931002
NCT Number: NCT06524414 **Sponsor Name:** Pfizer **Investigational Product:** PG4 (Multivalent Pneumococcal Conjugate Vaccine) **Phase of Development:** Phase 2 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety, tolerability, and immunogenicity of a new multivalent pneumococcal conjugate vaccine (PG4) in healthy infants when compared to currently licensed pneumococcal vaccines like 20vPnC (Prevnar 20®) or PCV15. **Secondary Objectives:** To evaluate if there is a difference in the safety and immune effects of the new PG4 vaccine when it is prepared in different types of solutions, and to measure the specific amount of antibodies produced after vaccination.

2.2 Background & Rationale To address the need for broader protection against pneumococcal disease in infants. *Streptococcus pneumoniae* has numerous serotypes that continuously change in prevalence, necessitating the development of multivalent conjugate vaccines that offer wider coverage against diseases like pneumonia and meningitis.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (20vPnC / Prevnar 20® and PCV15) **Blinding:** Double-blind **Randomization:** Randomized (Parallel Assignment into 4 groups)

3.2 Planned Enrollment & Duration **Target N\$:** 800 participants
Number of Sites: Multicenter (Global/US) **Estimated Study Start:** 25-Jul-2024 **Estimated Primary Completion:** 01-May-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female infants born at ≥ 36 weeks of gestation. 2. Infants who are exactly 2 months of age (42 to 98 days) at the time of consent. 3. Healthy infants determined by clinical assessment, including medical history and clinical judgment, to be eligible for the study.

4.2 Exclusion Criteria 1. History of severe adverse reaction associated with a vaccine and/or severe allergic reaction (e.g., anaphylaxis). 2. Major known congenital malformation, serious chronic disorder, or suspected immune system problems. 3. Previous vaccination with any licensed or investigational pneumococcal vaccine, or planned receipt through study participation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 4 doses of the study vaccine administered at 2, 4, 6, and 12 to 15 months of age. **Route of Administration:** Intramuscular injection (single shot into the left thigh muscle). **Duration of Treatment:** Total study participation lasts about 16 to 19 months (approximately 1.5 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard routine pediatric care that does not conflict with the study protocol. **Prohibited:** Previous/concurrent administration of any other licensed or investigational pneumococcal vaccines, and immunosuppressive therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days/Months)	Key Activities/Procedures
1	-14 to 0	Screening, Informed Consent, Medical History, Clinical Health Assessment
2	2	Vaccination
3	4	Vaccination
4	6	Vaccination
5	12-15	Vaccination, Safety Assessment
6	16-19	Interim Follow-ups, End of Study Evaluation

(Note: Participants will have a total of 6 in-person study clinic visits and 1 to 2 phone calls during the study duration)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability (measured by frequency of local/systemic side effects) and

Immunogenicity (amount of antibodies produced after vaccination). **Measurement Method:** Parent/guardian-reported side effects at visits/calls, and Central Laboratory Analysis of blood samples for pneumococcal antibody titers.

7.2 Secondary & Exploratory Endpoints Difference in the safety profile and immune response (antibody generation) between two different liquid mixture solutions of the PG4 vaccine.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Parent(s) or legal guardian(s) are prompted to report any unintentional or unexpected reactions (side effects) at all clinic visits and via phone calls. **Serious Adverse Events (SAEs):** Continuously monitored and evaluated over the 16-19 month study duration. **Data Monitoring Committee (DMC):** Internal/External safety monitoring board established by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized infants receiving at least one dose); Per-Protocol (PP) / Evaluable Immunogenicity population for antibody responses. **Sample Size Justification:** A target of 800 participants allocated sequentially across 4 groups ensures adequate statistical power to compare PG4 formulations against standard comparators (20vPnC/PCV15). **Handling of Missing Data:** Standard methodologies (e.g., Multiple Imputation) as dictated by the study's SAP for vaccine clinical trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine monitoring of clinical sites to ensure accurate AE reporting, adherence to vaccination schedules, and proper blood sample handling. **Audit Trail:** Strict documentation of parental informed consent, eCRF logging, and laboratory sample chain-of-custody.

11. Study Identifiers & Governance

Primary ID: C4931002 **Registry Number:** NCT06524414 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About How a New Pneumococcal Vaccine Works in Infants **Official Regulatory Title:** A Phase 2, Randomized, Double-Blind Trial to Evaluate the

12. Clinical Framework (Adapted for Pneumococcal
Vaccine Study)

Primary Condition: Pneumococcal Disease **Diagnosis Threshold:** N/A
(Healthy infants requiring prophylactic vaccination) **Medical
Methodology:** Multivalent Pneumococcal Conjugate Vaccine
administration **Optimization Target:** Maximum immunogenicity
(antibody response) with a highly tolerable safety profile

13. Study Procedures & Methodology

Management Pathway: Standard pediatric immunization pathway
Education Protocol (HCP): Site initiation and training on strict
blinding, vaccine preparation, and intramuscular administration
Education Protocol (Patient): Guardian education on recognizing
and reporting local/systemic adverse events **Quality Audit
Mechanism:** Guardian compliance tracking for all 6 clinic visits,
3 blood draws, and follow-up phone calls

14. Observation & Follow-Up Schedule

Total Duration: 16 to 19 months **Onsite Visit Cadence:** Months 2,
4, 6, and 12-15 (Total of 6 visits) **Remote Monitoring Frequency:**
1 to 2 follow-up phone calls **Checklist Review Interval:** Safety
side-effect review at every study contact

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				